SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 40%

QUESTIONS: 10 **Total Marks:50**

Annexure A

Coding challenge

Code of Conduct:

I S.Karthikeyan (192419048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
S.karthikeyan

Annexure A
Coding challenge

1. Write a Python script that generates a star topology diagram (ASCII or graphical) given n hosts and a central switch. The script must print each host label, the switch name, and the connecting Cat6 UTP cable length.
2. Code a function `validate_segment(length_m)` that returns True if the UTP segment is within IEEE 802.3 limits (≤100 m) and False with an error message otherwise. Include at least 3 test cases.
3. Write a script that simulates CSMA/CD on a star topology with 4 hosts. Log every collision detection event and the back-off interval to a file named `csma_log.txt`.
4. Using Python's `socket` library, write a client-server program that models two hosts communicating through a star switch. Demonstrate message forwarding by printing the switch's MAC address table after each frame transmission.
5. Create a class `StarSwitch` with methods `add_port(host_id)`, `remove_port(host_id)`, and `broadcast(frame)`. Broadcast must send a frame to all ports except the source and log each delivery.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Understanding	15%	Fully understands the problem and requirements; no clarification needed	Understands most requirements with minor clarification	Partial understanding; misses some requirements	Poor understanding of the problem
Algorithm Design	20%	Efficient, optimal, and well-structured algorithm	Correct algorithm with minor inefficiencies	Basic approach but not optimal	Incorrect or no clear algorithm
Code Quality	15%	Clean, well-organized, readable, and properly formatted code	Mostly clean code with minor issues	Code is somewhat unclear or inconsistent	Poorly written, unreadable code
Functionality	20%	Code works perfectly for all test cases	Works for most test cases	Works for limited cases	Does not run or fails major cases
Error Handling	10%	Handles all edge cases and errors effectively	Handles some edge cases	Limited error handling	No error handling
Efficiency	10%	Highly optimized in time and space complexity	Reasonably efficient	Some inefficiencies	Highly inefficient
Documentation & Comments	5%	Clear and meaningful comments and documentation	Adequate comments	Few or unclear comments	No comments
Testing & Debugging	5%	Thoroughly tested with multiple cases; no bugs	Tested with some cases	Minimal testing	No testing done

QUESTION 1:

```python
# Program to create a simple Star Topology

num_hosts = int(input("Enter the number of hosts: "))

switch = "Switch"

print("\n--- Star Topology ---")
print("Central Device:", switch)

for i in range(1, num_hosts + 1):
    host_name = "Host" + str(i)

    cable_length = float(
        input("Enter Cat6 cable length for " + host_name + " (in meters): ")
    )

    print(host_name, "----", cable_length, "m ----", switch)

print("\nStar topology created successfully.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
--- Star Topology ---
Central Device: Switch
Enter Cat6 cable length for Host1 (in meters): 10
Host1 ---- 10.0 m ---- Switch
Enter Cat6 cable length for Host2 (in meters): 15
Host2 ---- 15.0 m ---- Switch
Enter Cat6 cable length for Host3 (in meters): 20
Host3 ---- 20.0 m ---- Switch
Enter Cat6 cable length for Host4 (in meters): 25
Host4 ---- 25.0 m ---- Switch

Star topology created successfully.
PS C:\Users\admin\Downloads>
```

QUESTION 2:

```python
def validate_segment(length_m):

    if length_m <= 100:
        return True
    else:
        return "Error: Cable length is more than 100 meters"


# Test cases

print(validate_segment(50))
print(validate_segment(100))
print(validate_segment(120))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Enter Cat6 cable length for Host2 (in meters): 15
Host2 ---- 15.0 m ---- Switch
Enter Cat6 cable length for Host3 (in meters): 20
Host3 ---- 20.0 m ---- Switch
Enter Cat6 cable length for Host4 (in meters): 25
```

QUESTION 3:

```python
    # Randomly simulate a collision
    collision = random.choice([True, False])

    if collision:
        print("Collision detected by", sender)

        backoff = random.randint(1, 5)

        print("Back-off time:", backoff, "seconds")

        log_file.write(
            sender + " - Collision detected - Back-off: "
            + str(backoff) + " seconds\n"
        )

        time.sleep(1)

    else:
        print(sender, "sent data successfully")

log_file.close()

print("\nCSMA/CD simulation completed.")
print("Collision details are saved in csma_log.txt")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Host3 is trying to send data
Host3 sent data successfully
Host1 is trying to send data
Host1 sent data successfully
Host4 is trying to send data
Collision detected by Host4
Back-off time: 3 seconds
Host3 is trying to send data
Host3 sent data successfully

CSMA/CD simulation completed.
Collision details are saved in csma_log.txt
PS C:\Users\admin\Downloads>
```

QUESTION 4:

```python
def host1():
    print("Host1 sent frame:", message)

    client.close()


# Start switch and Host1
switch_thread = threading.Thread(target=switch)
host_thread = threading.Thread(target=host1)

switch_thread.start()
host_thread.start()

switch_thread.join()
host_thread.join()

print("\nMAC Address Table after frame transmission:")
for host, mac in mac_table.items():
    print(host, "->", mac)

print("\nCommunication completed successfully.")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
  File "c:\Users\admin\Downloads\New folder (2)\question 4.py", line 15, in switch
    server.bind(("localhost", 5000))
    ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^
OSError: [WinError 10048] Only one usage of each socket address (protocol/network address/port) is normally permitted
Host1 sent frame: Hello Host2

MAC Address Table after frame transmission:
Host1 -> AA:AA:AA:AA:AA:01
Host2 -> BB:BB:BB:BB:BB:02

Communication completed successfully.
PS C:\Users\admin\Downloads>
```

QUESTION 5:

```python
class StarSwitch:
    def broadcast(self, frame):
                print("Frame delivered to", host)


# Create switch
switch = StarSwitch()

# Add hosts
switch.add_port("Host1")
switch.add_port("Host2")
switch.add_port("Host3")
switch.add_port("Host4")

# Create a frame
frame = {
    "source": "Host1",
    "message": "Hello everyone"
}

# Broadcast the frame
switch.broadcast(frame)

# Remove a host
switch.remove_port("Host4")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\admin\Downloads> & C:\Users\admin\AppData\Local\Programs\Python\Python314\python.exe "c:/Users/admin/Downloads/New folder (2)/question 5.py"
Host1 connected to switch
Host2 connected to switch
Host3 connected to switch
Host4 connected to switch

Broadcasting frame: Hello everyone
Frame delivered to Host2
Frame delivered to Host3
Frame delivered to Host4
Host4 removed from switch
PS C:\Users\admin\Downloads>
```